Individual Coursework Submission Form

Specialist Masters Programme

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Module Code: SMM799	
Module Title: Applied Research Project	
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Earnings-Call Sentiment and Stock-Price Reaction: A Comparative Evaluation of FinBERT and DeepSeek in the Technology Sector

This report is submitted as part of the requirements for the award of the

Master of Science in Business Analytics

In Bayes Business School, City St. Georges University of London.

YEAR OF SUBMISSION

September, 2025

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Abstract

Equity markets react not only to what firms report but to how managers discuss results and outlook on earnings calls. Prior research links call language and tone to immediate price responses and trading activity, highlighting the information role of textual cues beyond the numeric release (Price et al., 2012; Matsumoto, Pronk and Roelofsen, 2011; Mayew and Venkatachalam, 2012). Against the backdrop of outsized market influence from large technology firms and an AI-driven cycle in 2024–2025, understanding which language models produce the most decision-useful sentiment signal has clear practical relevance (Welsh, 2024; Navin, 2025; Lopez-Lira and Tang, 2024; Wu et al., 2023).

This study evaluates whether earnings-call paragraph-level sentiment aligns with short-window stock-price reactions and which approach, FinBERT (a domain-adapted transformer for financial text) or a general-purpose LLM (DeepSeek), delivers the more actionable summary. Paragraph segmentation is adopted to preserve granularity and faithfully capture local changes in tone across a call, while remaining compatible with transformer context limits; call-level scores are obtained by aggregating paragraph outputs. FinBERT is selected as a finance-specialised baseline given evidence that domain-specific pretraining improves sentiment classification in financial contexts, and is contrasted with a modern general LLM prompted for structured sentiment outputs (Araci, 2019; Malo et al., 2014; Loughran and McDonald, 2011; Lopez-Lira and Tang, 2024).

Methodologically, a standard event-study framework tests the association between call-level sentiment and short-horizon returns around earnings announcements; manual annotations provide a human benchmark for model agreement (MacKinlay, 1997). This analysis focuses on large-cap technology issuers, Apple, Amazon, NVIDIA, where earnings communication can catalyse meaningful index-level moves, and offers practical guidance on when model-derived sentiment is informative for post-announcement monitoring (Welsh, 2024; Navin, 2025). The study also documents design trade-offs, paragraph vs. coarser units, aggregation choices, and model prompting, that materially affect downstream conclusions (Matsumoto, Pronk and Roelofsen, 2011; Rennekamp, Sethuraman and Steenhoven, 2022).

Acknowledgement

I am extremely grateful to Dr Vali Asimit for his dedication in organising the collaboration with our industry partner and for the encouragement and guidance provided throughout the project.

I would also like to express my sincere gratitude to Dr Ahmad Abu-Khazneh and Dr Jan Rock of the Citibank Innovation Lab for their technical and business guidance, as well as their timely feedback and insights at key stages. A heartfelt thank-you to our business partner for the opportunity to work on a live, high-impact problem.

My thanks also go to Dr Simone Santoni for incisive academic supervision and methodological direction. Finally, I am deeply thankful to my teammates for their excellent cooperation, diligence, and resilience in overcoming every obstacle during the process.

Disclaimer

This report was produced for academic assessment at Bayes Business School. The views expressed are those of the author and do not necessarily reflect those of Citibank or City St George's, University of London. No confidential or proprietary information from the industry partner has been used.

This document is for educational purposes only and does not constitute investment advice, a recommendation, or an offer to buy or sell any security. The analysis relies on publicly available data and third-party tools/APIs and is provided "as is," without warranties of completeness or accuracy.

Model outputs, including those generated by large language models are probabilistic and may contain errors. Any market associations reported are associational, not predictive, and should not be used as the basis for trading decisions without independent verification.

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Chapter 1

Introduction

Earnings days compress accounting results, guidance, and executive tone into a single market event. Prices often react within hours not only to what is reported but to how it is said. Prior research shows that the linguistic tone of earnings conference calls is associated with abnormal returns and trading volume, alongside the classic effects of surprises and guidance; vocal affect during calls is informative as well. These findings motivate careful, human-aligned measurement of language on the day results are released (Mayew and Venkatachalam, 2012).

This question is especially relevant in today's concentrated market. A handful of mega-cap technology firms (including Apple, Amazon, and NVIDIA) now carry unusually large weights in major indices, so their commentary can influence benchmarks on its own. Recent reporting places the top ten U.S. stocks at roughly a third to two-fifths of S&P 500 market capitalisation, underscoring why the tone of these firms matters beyond single-name analysis (Welsh, 2024; Navin, 2025).

The analysis focuses on Apple, Amazon, and NVIDIA and uses the most recent eight earnings-call transcripts for each firm. Transcripts are cleaned and segmented into coherent paragraphs (the unit at which readers perceive tone in prepared remarks and Q&A). Two independent sentiment signals are produced on the same paragraphs: *"FinBERT"*, a finance-tuned BERT variant trained on domain corpora (e.g., Financial PhraseBank), and a general large language model, *"DeepSeek"*, prompted to assign five ordered labels *{Strong Negative, Slight Negative, Neutral, Slight Positive, Strong Positive}*. The full paragraph set for each company was manually annotated on the same 5-bucket scale (Apple n=428; Amazon n=143; NVIDIA n=201) to anchor evaluation. Labels are mapped to a numeric scale *(−2…+2)*, averaged by quarter, and compared with each stock's *±2-day* move around the earnings date to test short-window association. Inputs and scaling are held constant across firms so that differences reflect model behaviour, not data handling (Araci, 2019; ProsusAI, 2020a; ProsusAI, 2020b).

This design addresses a practical measurement problem: which model yields a sentiment signal that is both faithful to human judgement and useful around earnings?

It compares how FinBERT and the LLM behave at the paragraph level, assesses which aligns more closely with human labels, examines which quarter-level series better lines up with market direction, and turns the evidence into deployable practices for investor relations, risk monitoring, and portfolio analytics (Lopez-Lira and Tang, 2024; Wu et al., 2023).

Chapter 2

Literature Review

Research on language and markets shows that what managers say and how they say it around earnings matters for prices. Early studies using dictionary methods documented that the tone of disclosures influences investor reaction (e.g., earnings press releases), establishing that language carries incremental information beyond numeric surprises (Henry, 2008; Davis, Piger and Sedor, 2012). Follow-on work focused on conference calls, finding that the linguistic tone of call transcripts predicts abnormal returns and trading volume, and can even dominate the information in earnings surprises over subsequent weeks (Price et al., 2012). Beyond word choice, vocal affect in the audio, the emotional state conveyed by executives' voices, also relates to returns, underscoring that investors react to both content and delivery (Mayew and Venkatachalam, 2012).

A parallel literature studies which parts of the call matter. Prepared remarks are scripted; Q&A is more spontaneous and often more informative. Large-sample evidence shows both segments add information, with multiple studies reporting that Q&A tone helps explain market responses and improves analysts' forecasts relative to prepared remarks alone (Matsumoto, Pronk and Roelofsen, 2011; Rennekamp, Sethuraman and Steenhoven, 2022). This motivates paragraph-level analysis that keeps local context from both sections instead of collapsing the call to a single document score (Matsumoto, Pronk and Roelofsen, 2011).

Methodologically, event studies provide the backbone for measuring short-window reactions. The *±2-trading-day window* used sits within the standard design advocated by MacKinlay and others, who show that surrounding prices can be used to infer an event's impact when markets incorporate information quickly. While many studies estimate abnormal returns, this analysis uses raw ±2-day returns as a descriptive diagnostic; robustness is reported in Methodology. (Campbell, Lo and MacKinlay, 1997).

On the measurement side, finance moved from dictionary counts to machine-learned classifiers. Loughran and McDonald's domain word lists corrected general dictionaries that mislabelled neutral financial terms as negative, a milestone for early textual finance

(Loughran and McDonald, 2011). With transfer learning, FinBERT fine-tunes BERT on financial corpora, typically outputting positive/neutral/negative labels and achieving state-of-the-art results on domain datasets such as Financial PhraseBank. FinBERT is widely used as a finance-specific baseline because it embeds domain vocabulary and reduces the need for large labelled sets (Araci, 2019; ProsusAI, 2020a; ProsusAI, 2020b).

The most recent wave puts large language models (LLMs) at the centre. Domain LLMs such as BloombergGPT demonstrate strong performance across finance tasks, while academic and central-bank studies report that general LLMs can extract structured facts from news and predict returns from headlines, often outperforming traditional methods (Wu et al., 2023; Lopez-Lira and Tang, 2024). Comparative evaluations show GPT-class models can rival or exceed domain-specific models on several financial text tasks, suggesting that general reasoning ability plus instruction-following can offset narrower domain pre-training in some settings (Lopez-Lira and Tang, 2024; Wu et al., 2023). These developments raise a practical question for earnings analysis: do domain models or general LLMs better reproduce human judgement and market-relevant tone?

Two gaps in the literature shape the design. First, most prior work reports a single text signal per call, whereas practitioners experience calls paragraph by paragraph as tone shifts across topics and speakers. A paragraph-level unit respects this structure and aligns with evidence that Q&A carries incremental information (Matsumoto, Pronk and Roelofsen, 2011). Second, few studies provide a human-anchored, head-to-head comparison of a domain classifier (FinBERT) and a general LLM on the same text, then trace how paragraph-level errors aggregate to quarter-level signals and short-window market alignment within a common event-study frame. The present study addresses that gap by using manual five-bucket annotations, a shared five-class schema for both models, an ordinal mapping to −2…+2 for aggregation, and a consistent ±2-day alignment.

Finally, this review clarifies why the analysis centres on Apple, Amazon, and NVIDIA. A concentrated U.S. equity market means a handful of mega-caps carry outsize index weights; their calls can move benchmarks and propagate sentiment system-wide. Studying eight recent quarters for each issuer provides a contemporaneous cross-

section through changing macro conditions, allowing a like-for-like comparison of model behaviour and market linkage (Welsh, 2024; Navin, 2025).

In summary, the literature establishes that tone in earnings disclosures matters, that Q&A adds incremental information, that event-study windows are the standard tool for short-run reactions, and that modern NLP offers competing paths, domain-specific (FinBERT) and LLMs (DeepSeek), for measuring sentiment. Building on these strands, the study contributes a human-anchored, paragraph-level comparison of FinBERT and an LLM across three bellwether firms, and links model behaviour to short-window price responses in a reproducible way (Araci, 2019; Loughran and McDonald, 2011; Lopez-Lira and Tang, 2024; Wu et al., 2023).

Chapter 3

Methodology

This study compares how language in earnings calls maps to human judgement and short-horizon price moves through a two-phase pipeline: Dataset Preparation and Analysis Methods. Phase two constructs dual paragraph-level sentiment signals *(FinBERT, DeepSeek)*, evaluates them against *manual annotations*, and aligns quarterly aggregates with a *±2 trading-day market reaction*. The scope spans recent quarterly calls for *Amazon, Apple, and NVIDIA*.

3.1 Dataset Preparation

3.1.1 Scope & Events

The corpus covers recent Amazon, Apple, and NVIDIA earnings calls, indexed by fiscal quarter *(e.g., Q1_2025)* so modelling can respect how results/guidance are consumed. Earnings dates are fetched programmatically via *"yfinance"* (Ticker.get_earnings_dates), then used to anchor a symmetric event window: *close-to-close T−2 to T+2 trading days*. This yields a uniform market-reaction series per quarter, kept as a merge key for later model comparison (averaged language → market bin). Earnings events are labelled by quarter using a one-quarter backshift so the market reaction (at the report date) lines up with the quarter whose results are being discussed. This "*−1 mapping"* is applied consistently for Apple, Amazon, and NVIDIA; sensitivity to an unshifted mapping is discussed in Limitations. The code implements this consistently across the three tickers, using "*yfinance"* for calendars and history, and *pandas/numpy* for date handling and returns.

3.1.2 Text Acquisition & Cleaning

(a) Source discovery. Transcripts are taken from the canonical *Motley Fool pages* to ensure coverage consistency across quarters/issuers. Pages are loaded with *Selenium* (Chrome via webdriver-manager), cookie dialogs dismissed, and the article body captured with resilient XPath fallbacks; only visible *"//p"* paragraph nodes are retained. This avoids navigation/promotional content and stabilizes scraping across page variants.

(b) Scraping. The scraper scrolls to trigger lazy content, then extracts only "*<p>*" within the article container (relative .//p to prevent DOM wandering). Extraction is deliberately conservative so later filters do not have to carry an excessive burden.

(c) Cleaning. Before filtering, a custom *demojibake* repair is applied (latin-1 → utf-8 salvage plus html.unescape) to fix *"â€"/â€™"* artifacts seen in scraped text. Targeted regex lists remove boilerplate: site-wide footer/promo strings ("All earnings call transcripts", "The Motley Fool has positions in and recommends"), phone numbers, replay instructions, and other nav/junk; standard legal blocks ("forward-looking", "safe harbor", "disclaimer") are dropped so non-substantive language does not inflate Neutral/Negative. Newlines are normalized and multi-breaks collapse. Cleaned quarters are saved as *UTF-8 .txt* for a reproducible baseline.

3.1.3 Paragraphization

Transcripts are segmented into analysis units that match how tone appears in practice: short, coherent paragraph blocks that include prepared remarks and Q&A. A strict regex header pattern detects *speaker/title lines (e.g., "Name -- Title")* and opens a new block when encountered; bracketed operator notes […] are stripped; fragments below a small character threshold are discarded to limit noise. Missing speaker/title on continuation lines are forward-filled with *pandas* so every block preserve attribution. The standardized schema, {*quarter, speaker, title, content*}, is exported as per-company CSVs, providing stable paragraph IDs for cross-model comparison.

3.2 Analysis Methods

Analysis constructs two independent sentiment signals on the same paragraphs, benchmarks them against human annotations, and then aligns quarterly aggregates to market reaction.

3.2.1 Sentiment Models (two independent signals)

(a) FinBERT pipeline. We use *Hugging Face Transformers* with the *ProsusAI/finbert* checkpoint to obtain paragraph-level labels and probabilities via the standard pipeline *"sentiment-analysis"*. Because BERT models cap sequence length (~512 tokens), long paragraphs are split into *≈200-word chunks*, scored, and collapsed to a single paragraph label by majority vote (ties broken by the maximum-confidence chunk). The coarse positive/neutral/negative result is then mapped to a five-bucket schema by applying a confidence threshold *(>0.85 ⇒ Strong±; otherwise, Slight±)*, with *Neutral* retained as given. Outputs are stored as 5-bucket labels *{Strong−, Slight−, Neutral, Slight+, Strong+}*.

(b) DeepSeek classifier. In parallel, an *OpenAI-compatible* client prompts DeepSeek to assign calibrated probabilities over the same 5 buckets for each paragraph; the response is parsed into columns such as "*LLM_pct_slightly_positive*", etc., and the argmax label is kept alongside the probability vector. We store both 5-bucket labels and a 3-class collapse for robustness diagnostics.

(c) Label mapping & confidence. Manual, FinBERT, and DeepSeek labels are aligned onto a numeric score *(−2…+2)* to enable averaging, rank tests, and market comparison on a common ordinal backbone. Probability outputs (from both models) are retained to audit low-confidence assignments and inspect intensity compression (e.g., Strong → Slight).

(d) Human ground truth. For each company, the full paragraph set used in evaluation was manually annotated on the same 5-bucket scale *(Apple n=428; Amazon n=143; NVIDIA n=201)*. These human labels are used solely as ground truth for per-class diagnostics and quarter-level stability checks and are not used for model training.

(e) Metrics (method → measure → visual). Using *scikit-learn*, we report *Accuracy, Macro-F1,* and *Weighted-F1;* then render two *Confusion Matrices* per model: raw counts (volume) and row-normalized (recall by true class). The evaluation frame enforces a fixed class order to keep axes consistent across companies, and the same code template is reused for Apple/Amazon/NVIDIA to preserve comparability. Visuals are plain *Matplotlib/Seaborn* for legible heatmaps, with the full classification report captured to tidy tables for the appendix.

3.2.2 Market-Reaction Construction

(a) Event window. For each earnings date, we compute a close-to-close return from *"T−2 to T+2"* trading days using *yfinance* price history, with trading-day selection performed directly on the index to avoid calendar gaps. This symmetric window balances *"surprise absorption"* with day-of microstructure noise.

(b) Binning. Returns are bucketed into five ordered categories to mirror the sentiment scale *(e.g., Strong Positive if > 5%, Neutral if within ±1%).* This aligns the market series to the same ordinal space as the models/humans, enabling like-for-like aggregation and plotting.

(c) Linking. Quarter-average paragraph scores (Manual, FinBERT, DeepSeek) are joined to the quarter's market bucket. We visualize these two ways: (i) a *Plotly price timeline* with colour markers for each earnings reaction (EPS surprise in the hover), and (ii) *a comparison panel* overlaying the market line atop bar charts of the three quarter-level averages.

3.2.3 Aggregation & Alignment

Paragraph labels from all sources are mapped to (−2…+2) and averaged within a quarter so calls with more paragraphs don't dominate by construction, yet still contribute precision. Per-company panels display the price series with event markers and the model/human quarter averages against the market line, making co-movement

and divergences easy to read without statistical over-claim. Where appropriate, we also hold a 3-class collapse and probability-aware summaries to check robustness.

3.3 Limitations

Several factors limit how far these results can be taken. The data are Neutral-heavy, with few negatives, which can make Accuracy look better than it is and pull down Macro-F1. The analysis uses transcripts only; news and macro events around the call can move prices too. Market reaction is measured as raw ±2-day returns, so findings are descriptive, not causal, and may change with a different window. Earnings events are aligned to quarters using a −1 backshift (the report date mapped to the prior content quarter); other mappings are possible. To fit model context limits, long paragraphs are chunked, which can soften extremes, and model/prompt settings can bias results toward "slight" classes. Scope is narrow, two models, three firms, eight quarters, so generalisability is limited. Manual labels may include annotator bias; inter-annotator agreement was not recorded. DeepSeek versioning/temperature and scrape/clean quirks can add variance. Finally, quarter-level averaging can hide speaker or topic effects within a call.

Chapter 4

Results & Analysis

This section maintains a single thread from paragraph-level behaviour to quarter-level outcomes. For each company, the transcript sample is first characterised to clarify what the models were asked to classify; next, the alignment of FinBERT and DeepSeek with human judgements is evaluated at the paragraph level (identifying systematic over- or under-calls); finally, paragraph scores are aggregated to the quarter level and compared with the stock's ±2-day move around the earnings date, with agreements and divergences interpreted in business terms.

Table 4.1 summarises paragraph-level performance for FinBERT and the LLM across all three companies; detailed confusion matrices and quarter overlays follow in 4.1–4.3.

Company	Model	Accuracy	Macro-F1	Weighted-F1
Amazon	FinBERT	0.678	0.452	0.643
Amazon	LLM (DeepSeek)	0.685	0.356	0.658
Apple	FinBERT	0.743	0.416	0.744
Apple	LLM (DeepSeek)	0.715	0.376	0.698
NVIDIA	FinBERT	0.577	0.310	0.490
NVIDIA	LLM (DeepSeek)	0.672	0.651	0.648

4.1 Amazon - conservative readings versus nuanced readings, and a single decisive miss

Across eight recent quarters, the sample contains 143 Amazon paragraphs. The human ground truth concentrates in *Neutral* and *Slightly Positive*; *Strong Negatives* are absent and *Strong Positives* are few (ten paragraphs). That class shape matters: any model can

look steady by leaning on *Neutral*, but to be useful it must still recognise the modest optimism that actually carries most paragraphs.

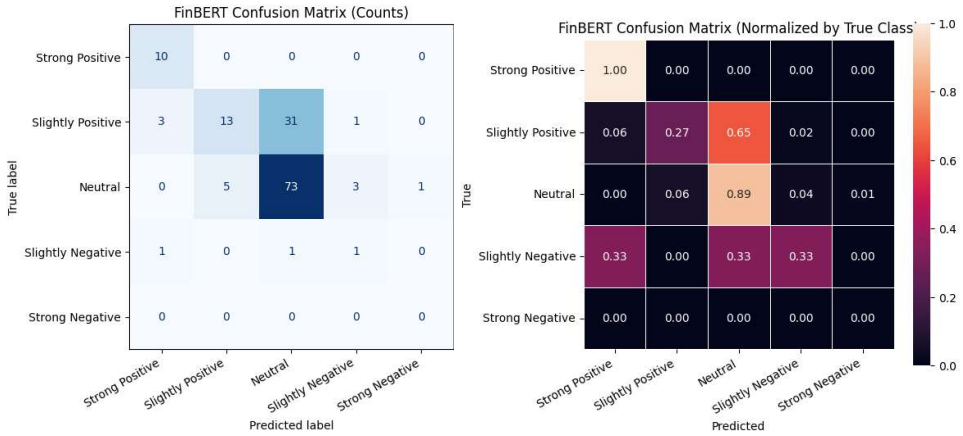

Figure 4.1-A: ***Amazon: FinBERT confusion matrices (left: counts; right: row-normalised).*** *n = 143; class order = {Strong Positive, Slightly Positive, Neutral, Slightly Negative, Strong Negative}. Highlights: perfect recall for Strong Positive (10/10); Neutral recall ≈ 0.89; Slightly Positive recall ≈ 0.27 with ≈ 65% reassigned to Neutral; no Strong Negative in truth. Overall pattern: conservative, excellent on Neutral and clear positives, but mutes mild optimism.*

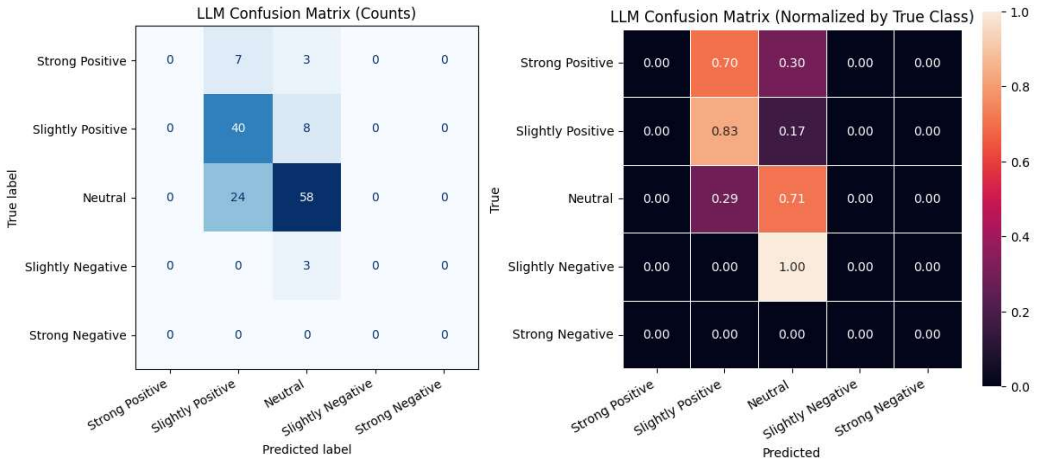

Figure 4.1-B: ***Amazon: LLM (DeepSeek) confusion matrices (left: counts; right: row-normalised).***

n = 143; class order = {Strong Positive, Slightly Positive, Neutral, Slightly Negative, Strong Negative}. Highlights: model effectively uses two labels (Neutral/Slightly Positive); Slight+ recall ≈ 0.83; Neutral recall ≈ 0.71 with ≈0.29 shifted to Slight+; Strong+ softened (0% predicted Strong+, ≈70% → Slight+, ≈30% → Neutral); Slight− mapped to Neutral (recall 0). Overall pattern: momentum-sensitive, captures mild positives but compresses extremes and under-detects slight negatives.

On this distribution FinBERT behaves like a careful analyst. It matches the human label in roughly 68% of cases and posts a *macro-F1 (~0.45) (Table 4.1)*, anchored by *strong Neutral recall (~0.89)* and *perfect Strong Positive recall (10/10) (Fig 4.1-A).* The price of that caution is systematic: about 65% of human Slightly Positive paragraphs are flattened to Neutral, so small signals of improvement are often muted in FinBERT's output. DeepSeek behaves differently. It almost entirely uses Neutral and Slightly Positive (no Strong+, no negatives), yet lands on a very similar overall *accuracy (~68.5%) (Table 4.1)* because those two classes dominate the corpus. Crucially, it recovers the *Slightly Positive recall (~0.83)) (Fig 4.1-B)* that FinBERT tends to wash out, while compressing intensity (human Strong+ becomes DeepSeek Slight+). In short: FinBERT captures clear wins; the DeepSeek surfaces soft wins, and those are exactly what this dataset contains.

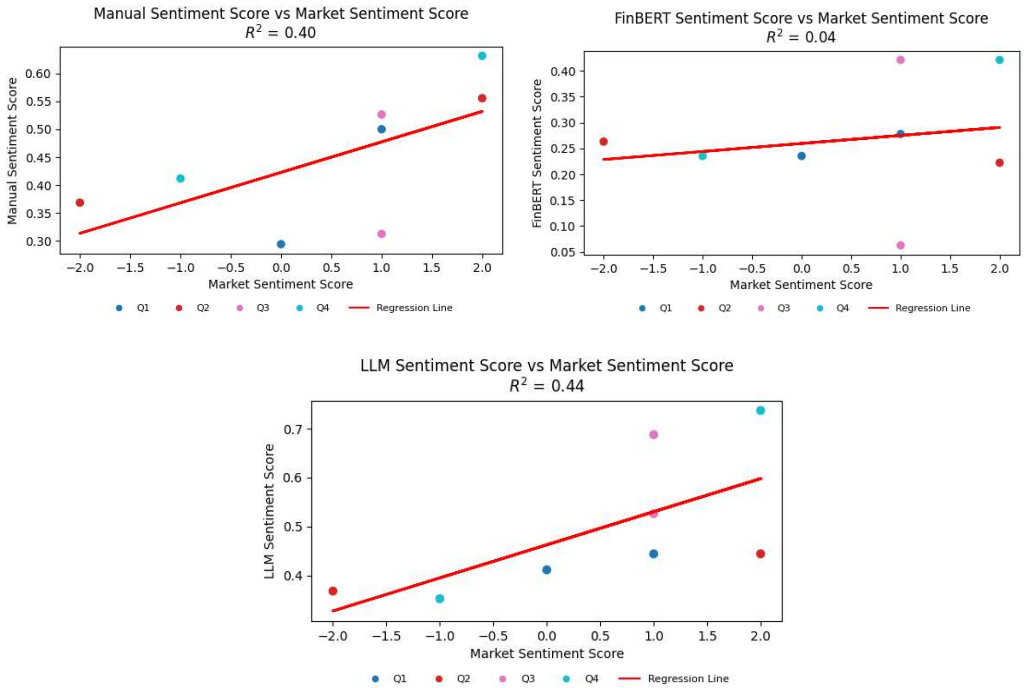

Figure 4.1-C: ***Amazon: Quarter sentiment vs market "score" (scatter with OLS fit).***
Left: Manual (R^2 = 0.40) · Centre: FinBERT (R^2 = 0.04) · Right: LLM (R^2 = 0.44). Takeaway: LLM shows the strongest co-movement; FinBERT is near-flat; Manual is moderate.

Aggregating to the quarter level makes that trade-off visible against the market. When each paragraph to a numeric score (−2…+2) and average within quarter, the DeepSeek's quarterly sentiment moves with the stock's event-window reaction more than the alternatives; a simple regression of quarter average sentiment to the market's own five-bucket reaction yields the best fit for the DeepSeek and a near-flat line for FinBERT *(Fig 4.1-C)*. In the Amazon scatter, quarters that rallied sit in the upper-right with higher DeepSeek averages, while FinBERT's quarter means barely respond because its paragraph outputs sit close to Neutral even in upbeat calls.

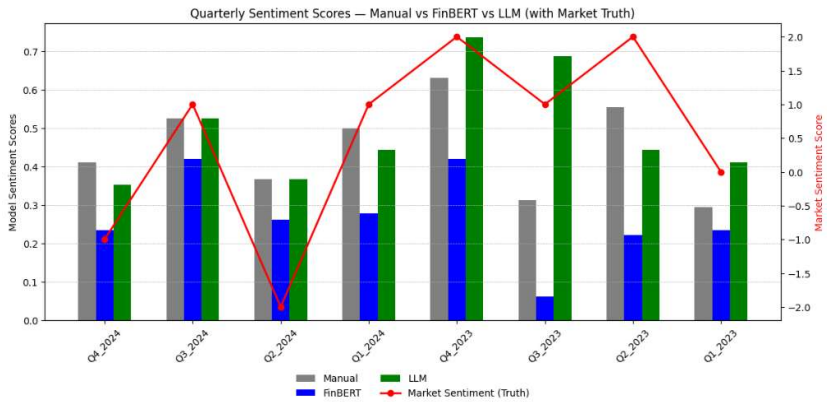

Figure 4.1-D: ***Amazon: Quarter-level sentiment overlay.***

Bars = quarter means of paragraph labels (−2…+2) for Manual, FinBERT, and LLM; red line = ±2-day market reaction (5-bucket scale). LLM co-moves with the market (notably Q4-2023, Q2-2023), while FinBERT stays compressed near Neutral; Q2-2024 is the key divergence, positive tone, negative price.

There is one quarter that all methods fail to anticipate: *Q2-2024*, when Amazon's stock fell sharply in the ±2-day window while all three sentiment sources stayed mildly positive *(Fig 4.1-D)*. Contemporary reporting ties that drop to a revenue miss and a softer-than-expected outlook, which dominated investors' response even though the call's language did not sound negative. In other words, guidance and expectations, not tone, drove the price that week. That exception is instructive rather than disqualifying: it shows why text-only signals should be used as context, especially when the market is trading guidance more than narrative (Zaveri and Sophia, 2024).

For business readers, the Amazon story is therefore cohesive. FinBERT gives a conservative baseline that rarely over-reads positivity and will reliably surface any obviously upbeat passage; DeepSeek provides sensitivity to incremental optimism so that quarter-to-quarter uplift shows up rather than being neutralised. The quarter-level overlay confirms that when Amazon rallies after results, the DeepSeek's average usually rises with it, whereas FinBERT stays subdued; and when guidance overwhelms tone (as in *Q2-2024*), all text signals can be on the wrong side of the move.

4.2 Apple - a neutral core with edge-class nuance, and divergences that teach the right lesson

Apple's dataset is substantially larger, 428 paragraphs, and even more skewed to Neutral in the human ground truth (about 74% of all paragraphs), with *no Strong Negatives* at all. This is Apple's house style: measured, precise, and spare with extremes. That style accentuates the methodological question of whether a model can pick up slight positives and negatives without hallucinating.

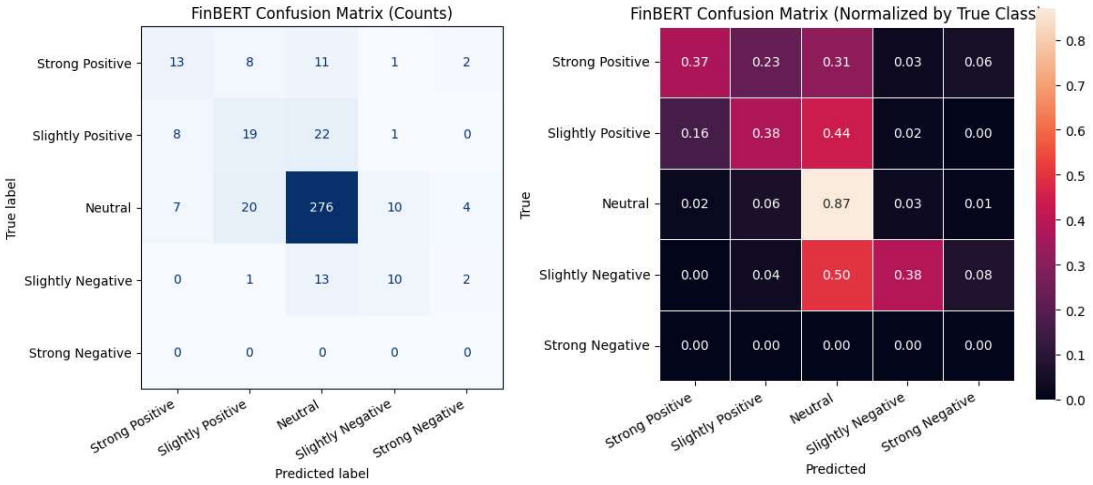

Figure 4.2-A: ***Apple: FinBERT confusion matrices (left: counts; right: row-normalised).*** *n = 428; class order = {Strong Positive, Slightly Positive, Neutral, Slightly Negative, Strong Negative}. Highlights: Neutral recall ≈ 0.87; Slightly Negative recall ≈ 0.38 (a key strength); Slightly Positive recall ≈ 0.38 with ≈ 0.44 shifted to Neutral; Strong Positive often down-rated (≈0.31 to Neutral, ≈0.23 to Slight+). No Strong Negative in the ground truth.*

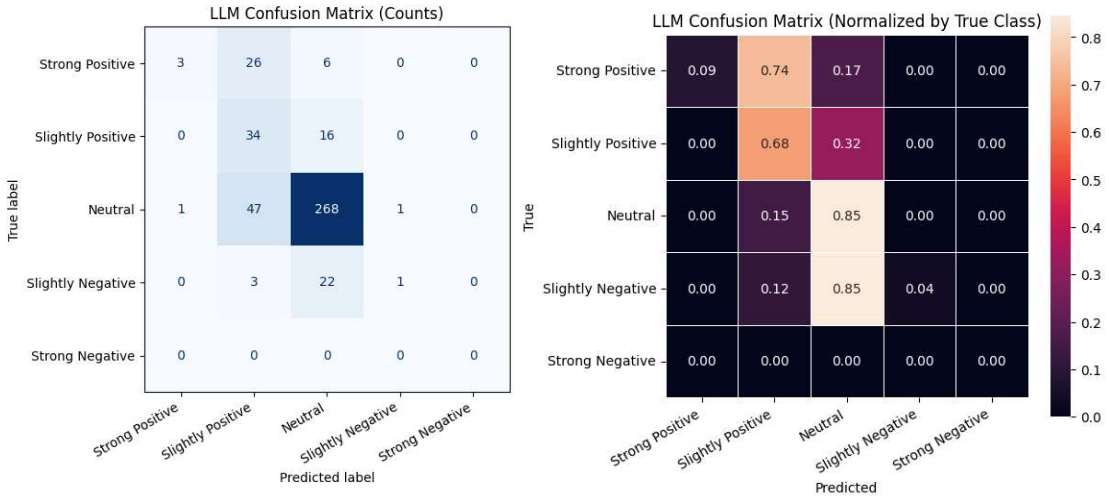

Figure 4.2-B: ***Apple: LLM (DeepSeek) confusion matrices (left: counts; right: row-normalised).***

n = 428; class order = {Strong Positive, Slightly Positive, Neutral, Slightly Negative, Strong Negative}. Highlights: model concentrates on Neutral/Slight+; Slight+ recall ≈ 0.68; Neutral recall ≈ 0.85 with ≈0.15 shifted to Slight+; Strong+ softened (≈0.74 → Slight+, ≈0.17 → Neutral, ≈0.09 kept as Strong+); Slight− mostly mapped to Neutral (≈0.85; ≈0.12 → Slight+). No Strong− in ground truth. Overall pattern: momentum-sensitive, captures mild positives, compresses extremes, under-detects slight negatives.

Against human labels, FinBERT offers the best overall balance here *Accuracy (~0.74) and Macro-F1 (~0.42) (Table 4.1)*. It is extremely reliable on *Neutral's recall (~0.87),* it recovers more Slightly Negative paragraphs than the DeepSeek, *recall (~0.38)* versus near-zero for the DeepSeek, and it preserves a material share of *Strong Positive recall (~0.37) (Fig 4.2-A)*. Where it softens is predictable: both Slightly Positive and Strong Positive are often nudged toward Neutral or down-rated to slight. DeepSeek sits just behind in top-line *accuracy (~0.72) (Table 4.1)* with a lower *macro-F1 (~0.38) (Fig 4.2-B),* but it excels at Apple's most common non-neutral class, Slightly Positive, recovering roughly two-thirds of those paragraphs. In other words, FinBERT is better for guarding the negative and keeping extremes honest, while the DeepSeek is better for capturing the modest optimism that Apple frequently communicates in cautious language.

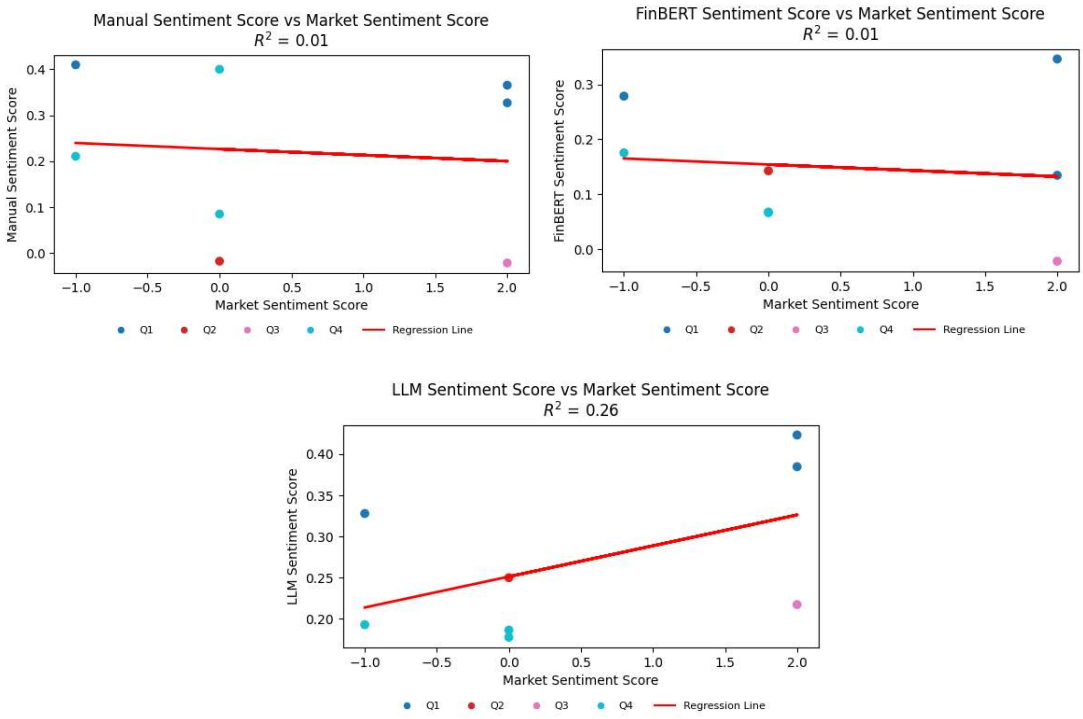

Figure 4.2-C ***Apple: Quarter sentiment vs market "score" (scatter with OLS fit).***
Left: Manual (R^2 = 0.01) · Centre: FinBERT (R^2 = 0.01) · Right: LLM (R^2 = 0.26). Takeaway: modest co-movement only for the LLM; Manual and FinBERT are effectively flat, consistent with Apple's neutral house style.

Quarter-level alignment reflects that division of labour. When the three paragraph signals are averaged by quarter and compared with Apple's ±2-day move, DeepSeek's quarterly sentiment shows a modest positive association with the market, whereas FinBERT and even the human averages exhibit little linear relationship *(Fig 4.2-C)*. That is not surprising: Apple's tone is mostly neutral-to-slightly-positive most quarters, so only a model that is willing to record those small upward shifts will show any co-movement with price.

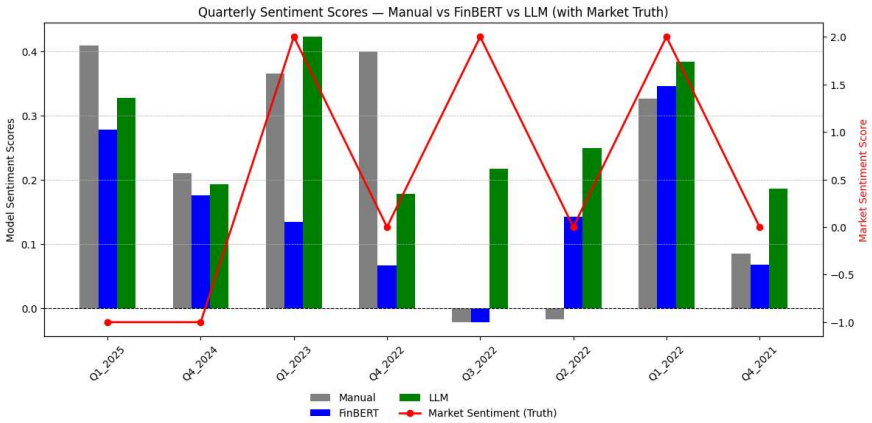

Figure 4.2-D ***Apple: Quarter-level sentiment overlay.***
Bars = quarter means of paragraph labels (−2…+2) for Manual, FinBERT, and LLM; red line = ±2-day market sentiment (5-bucket score). Pattern: the LLM captures small positive shifts (e.g., Q1-2023, Q1-2022), while Manual/FinBERT stay closer to Neutral; classic divergences show up in Q3-2022 (price up, tone muted) and Q4-2024/Q1-2025 (tone mildly positive, price flat/negative).

The quarterly bar-and-line chart *(Fig 4.2-D)* makes this concrete. Quarters such as *Q1-2022* and *Q1-2023* show a positive tone with positive market reaction; the models and the market tell the same story. There are quarters where tone and outcome diverge. In *Q3-2022*, for example, management's language reads cautious, yet the stock jumps, likely a case of beating a low expectations bar. And in *Q4-2024,* Apple beats but the stock dips after hours as investors focus on China demand and guidance texture; the transcript tone remains mildly positive, but macro and forward-looking concerns dominate that week's trade. These are precisely the quarters where a tone–price overlay adds value: it does not predict, but it flags where investors and management were talking past each other (Bary and Dawkins, 2024).

Taken together, Apple confirms the complementary strengths observed in Amazon. FinBERT is the instrument for baseline tone and risk vigilance, it will not miss the few negatives Apple does express and it will keep exuberance in check; DeepSeek supplies momentum sensitivity to capture incremental optimism that can explain why some quarters' narratives feel better to investors even when the words remain measured. The

mixed quarter-level correlation is the right warning label: tone alone is not the whole story in a short window, especially for a company that often communicates with deliberate understatement.

4.3 NVIDIA - exuberant language, human-aligned DeepSeek reads, and the "expectations" trap

NVIDIA's transcripts present the mirror image of Apple's: the eight-quarter sample contains 201 paragraphs with very few negatives and a large share of positives (e.g., 53 Strong Positives and 39 Slight Positives in the human labels). This is the vocabulary of the AI boom years, and it sets a high bar for any conservative model: if a classifier tends to default to Neutral, it will erase the exuberance that human readers clearly perceive.

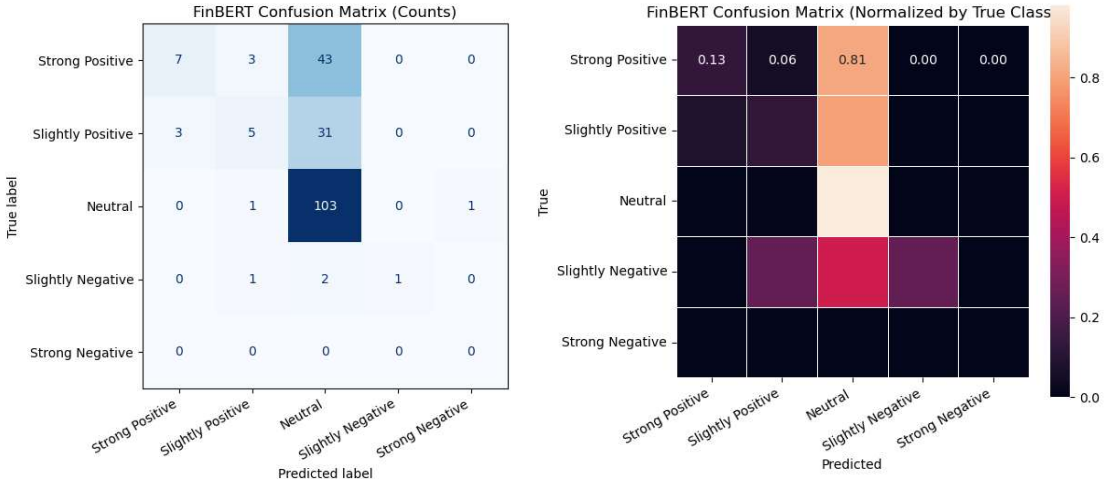

Figure 4.3-A ***NVIDIA: FinBERT confusion matrices (left: counts; right: row-normalised).***

n = 201; class order = {Strong Positive, Slightly Positive, Neutral, Slightly Negative, Strong Negative}. Highlights: Neutral recall ≈ 0.98; strong compression of positives, Strong+ → Neutral ≈ 0.81 (≈0.13 kept as Strong+); Slight+ recall ≈ 0.38 with ≈ 0.44 shifted to Neutral; negatives are sparse, with Slight− split across Neutral (0.50) and Slight− (0.38). Overall: very conservative, flattens NVIDIA's upbeat tone.

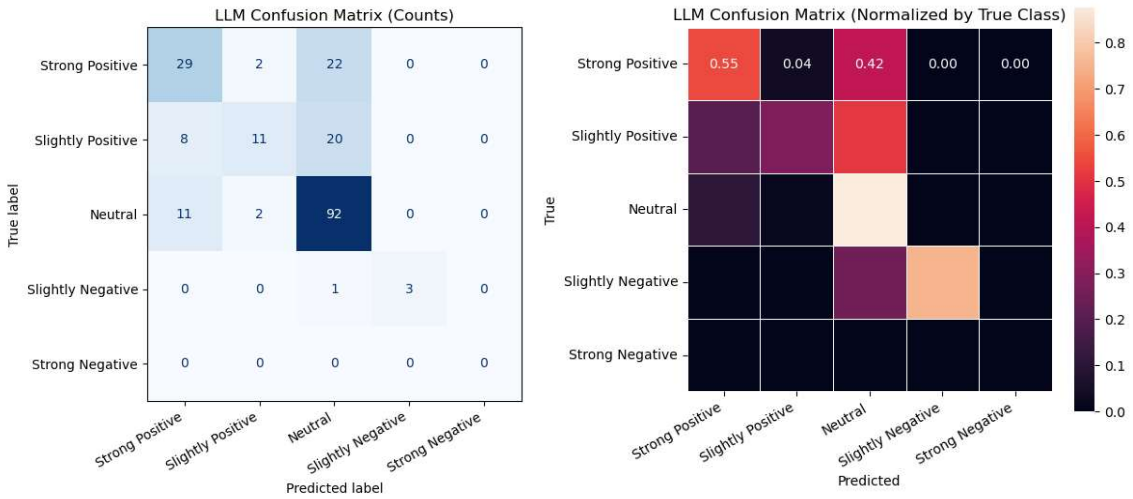

Figure 4.3-B ***NVIDIA: LLM (DeepSeek) confusion matrices (left: counts; right: row-normalised).***

n = 201; class order = {Strong Positive, Slightly Positive, Neutral, Slightly Negative, Strong Negative}. Highlights: Strong+ recall ≈ 0.55 (≈0.42 softened to Neutral); Neutral recall ≈ 0.88; Slight+ recall ≈ 0.28 with ≈0.51 shifted to Neutral; Slight− recall ≈ 0.75 on a very small base (n=4). No Strong− in the ground truth. Pattern: momentum-sensitive, recovers many Strong Positives while compressing mild positives toward Neutral.

That is exactly what the metrics show for FinBERT. At the paragraph level it delivers *accuracy (~58%)* but a *macro-F1 (~0.31) (Table 4.1)*, with *Neutral recall (~0.98)* and *recall on Strong Positive and Slight Positive down at (~0.13) (Fig 4.3-A)*. In practical terms, about eight in ten human-labelled Strong Positives are predicted Neutral by FinBERT, so the distinctive emotional register of NVIDIA's calls is largely flattened. DeepSeek reverses that pattern. Its *accuracy* rises to *(~0.67)*, *macro-F1* climbs materially *(~0.65) (Table 4.1),* and, most importantly, it recovers a majority of *Strong Positives recall (~55%)* while still keeping *Neutral recall high (~0.88) (Fig 4.3-B).* Both models struggle with negatives because there are almost none, but on the margin where it matters, capturing the positivity humans hear, DeepSeek is decisively closer to the ground truth.

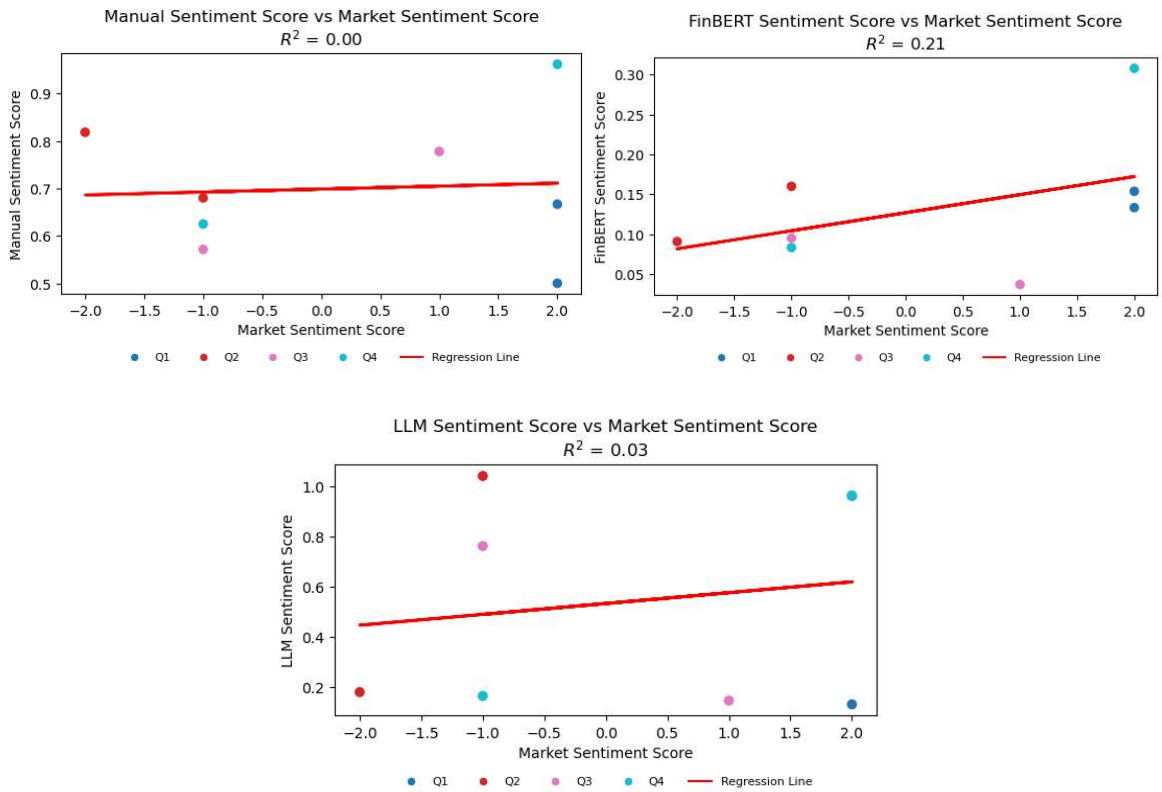

Figure 4.3-C ***NVIDIA: Quarter sentiment vs market "score" (scatter with OLS fit).***
Left: Manual (R^2 = 0.00) · Centre: FinBERT (R^2 = 0.21) · Right: LLM (R^2 = 0.03). Takeaway: all fits are weak; FinBERT's small edge comes from its compressed, near-neutral range rather than strong predictive power, expectations/valuation effects dominate.

When we roll those paragraph scores up to quarters and place them against the ±2-day move, the message becomes more subtle. Nearly every quarter's average sentiment is positive (DeepSeek and Manual bars stay above zero throughout), yet the stock has both strong up moves and meaningful pullbacks over the same window. Linear fits are accordingly weak for all methods; the apparent small edge for FinBERT in some regressions is largely an artefact of its compressed, near-neutral range rather than genuine predictive power *(Fig 4.3-C)*. The useful information in NVIDIA comes not from slope but from disconnects: quarters where management's language is

emphatically upbeat but the market sells on lofty expectations, valuation, or guidance nuance.

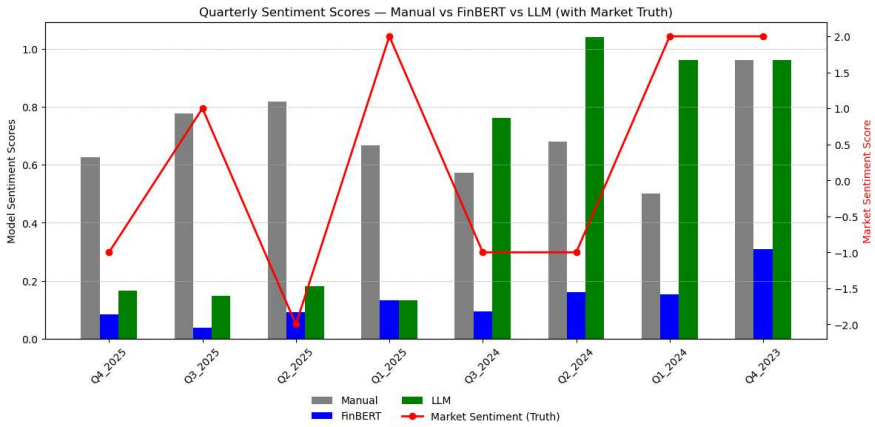

Figure 4.3-D ***NVIDIA: Quarter-level sentiment overlay.***
Bars = quarter means of paragraph labels (−2…+2) for Manual, FinBERT, and LLM; red line = ±2-day market sentiment (5-bucket score). Pattern: tone is consistently positive (Manual/LLM > 0 across quarters) while the market alternates between rallies and pullbacks; LLM reflects the uplift most, FinBERT stays compressed near Neutral. Notable divergence: Q2-2024, upbeat tone but negative market, consistent with expectations/valuation effects.

One illustrative case is *"Q2-2024"* (reported late August 2024) *(Fig 4.3-D)*. NVIDIA delivered record revenue again, and the tone of the call was celebratory; yet the stock fell after the print as investors judged that the results did not clear the "whisper" numbers and some took profits. In our bar-and-line plot this appears as high sentiment bars and a negative market bucket. That is the point of the overlay: it does not forecast the move, but it tells you which quarters demand further reading of guidance and Q&A because tone ≠ price (Prakash and Deighton, 2024).

For business users, NVIDIA highlights why two lenses are better than one. DeepSeek is the right instrument for registering enthusiasm the way humans do; it ensures that the tone of an AI-Supercycle call is not filed as Neutral. FinBERT stays valuable as a grounding reference: when DeepSeek's quarterly average is high but FinBERT remains close to zero, the positivity may be arriving in soft, contextual ways rather than in explicit claims, an informational nuance for Investor Relations when calibrating

message and expectations. And when the red line dips while bars remain high, the analysis correctly points teams to expectations management and guidance interpretation as the levers to examine first.

4.4 What the three stories have in common

Across Amazon, Apple, and NVIDIA, a consistent division of labour is evident. FinBERT functions as a conservative anchor, high recall on Neutral and reliable detection of soft negatives, whereas DeepSeek is momentum-sensitive, better at picking up mild positives but prone to compress extremes. These systematic asymmetries explain the quarter-level patterns and tone–price divergences (strongest co-movement for Amazon, modest for Apple, expectation-driven mismatches for NVIDIA). On this basis, the models are best used as complements; Chapter 6 operationalises this via a compact rule-set, an executive-first reporting design, and light governance.

Chapter 5

Discussion

Table 5.1 Quarter-level co-movement (R^2) by company and model

R^2 from OLS of quarter mean sentiment (−2…+2) on the market sentiment score (−2…+2) and on raw ±2-day return.

Company	Model	R^2 vs Market Score	R^2 vs Raw % Change
Amazon	Manual	0.40	0.35
Amazon	FinBERT	0.04	0.02
Amazon	LLM (DeepSeek)	0.44	0.38
Apple	Manual	0.01	0.00
Apple	FinBERT	0.01	0.01
Apple	LLM (DeepSeek)	0.26	0.22
NVIDIA	Manual	0.06	0.04
NVIDIA	FinBERT	0.21	0.17
NVIDIA	LLM (DeepSeek)	0.07	0.05

The three-company experiment reveals one consistent pattern with two complementary strengths *(Table 5.1)*. FinBERT behaves like a conservative anchor: it excels at Neutral paragraphs, preserves unmistakably upbeat passages, and persistently under-calls mild positivity. DeepSeek behaves like a momentum reader: it is quick to label Slightly Positive, often downgrades intensity (Strong+ → Slight+), and under-detects slight negatives. These systematic errors are not bugs; they are stable behaviours that explain every result we saw at the quarter level. On Amazon, FinBERT's conservatism flattened cautious optimism while DeepSeek surfaced it; on Apple, FinBERT retained the few negatives and kept exuberance in check while DeepSeek captured small uplifts; on

NVIDIA, where positivity dominates, DeepSeek aligned far better with human judgement by recovering most of the Strong Positives that FinBERT neutralised.

When each shines follows directly from those patterns. In neutral or risk-sensitive contexts, FinBERT is the better first lens: on Apple it achieved strong overall accuracy and uniquely recovered the small pool of Slightly Negative paragraphs that DeepSeek largely ignored, a property you want for early warning and compliance reads. In optimism-heavy or nuance-heavy contexts, DeepSeek adds the signal you would otherwise miss: on Amazon it captured the mild, widespread positivity that best mirrored quarter-to-quarter market direction; on NVIDIA it recovered a majority of Strong Positives and lifted quarter averages to match what humans actually heard in the AI-boom narrative. In short, FinBERT = precision and restraint; DeepSeek = sensitivity and narrative momentum.

Why tone→price fits are weak is also clear. A ±2-day window around earnings is driven as much by guidance, expectations, positioning, and the macro tape as by tone. It was observed "text-up, price-down" quarters that news coverage linked to guidance/expectations gaps rather than the language of the call: *Amazon Q2-2024* (revenue miss and softer guidance) (Zaveri and Sophia, 2024), *Apple Q4-2024* (beat, but investor focus on China and guidance texture) (Bary and Dawkins, 2024), and *NVIDIA Q2-2024* (beat, but "whisper numbers" were higher and profit-taking followed) (Prakash and Deighton, 2024). In each case our sentiment bars stayed positive, but the red line fell, exactly what you expect when ex-text information dominates the short window. These are not model failures; they are reminders that text is descriptive context, not a standalone trading rule.

How to read divergence quarters is the key managerial lesson. When DeepSeek/Human are positive but price is negative, treat this as an expectations/guidance flag: revisit the outlook language, Q&A caveats, or valuation context (e.g., NVIDIA's "sell-the-news" dips despite celebratory tone) (Prakash and Deighton, 2024). Conversely, price up with cautious tone suggests bar-clearing beats or a supportive macro tape (e.g., Apple instances where measured language coincided with a relief rally) (Bary and Dawkins, 2024). This framing turns tone–price disagreement into a shortlist for qualitative drill-down rather than a contradiction to explain away.

Implications for Investor Relations and strategy are practical and immediate. Use FinBERT as the risk-aware guardrail: it stabilises dashboards, aligns with the neutral core of corporate speech, and catches the scarce negatives (especially for Apple-like messaging). Use the LLM as the momentum gauge: it highlights incremental optimism that stakeholders notice and, on names like Amazon, often co-moves with the market's direction at the quarter level. Most importantly, monitor the gap between them. A large LLM-positive / FinBERT-neutral gap means the narrative is upbeat in flavour but light on explicit claims; IR teams can choose to clarify evidence or tamp expectations, and analysts know to check guidance detail. For NVIDIA-style narratives, that gap is the feature, not the bug: it signals that enthusiasm is being communicated implicitly, precisely where expectations management matters most.

The cross-company takeaway is therefore coherent and actionable. FinBERT and DeepSeek are not substitutes; they are complementing whose biases map cleanly onto different business needs. FinBERT preserves discipline and reduces false alarms; DeepSeek preserves nuance and reduces missed positives. Tone alone will not predict every ±2-day move, but used together, and read through the lens of divergence, they deliver a faithful, human-aligned account of narrative tone and a reliable set of investigation flags when markets react to something other than words. This synthesis sets up the recommendations that follow: deploy both models, report them side-by-side, and encode simple rules that turn their agreement and disagreement into decisions.

Chapter 6

Recommendations

The evidence across Amazon, Apple, and NVIDIA supports a complementary deployment of the two models. FinBERT should be used as the conservative anchor that reliably captures Neutral tone and soft negatives, while DeepSeek should be used to surface slight positives and narrative momentum. This division of labour yields a more stable and actionable signal than either model alone.

Deployment rule-set. Apply a simple, auditable ensemble: treat any FinBERT Slight−/Strong− as Negative; accept Strong+ only when FinBERT = Strong+ and DeepSeek is non-negative; accept Slight+ when the DeepSeek = Slight+ and FinBERT is non-negative; route remaining disagreements to a short human review queue. This preserves FinBERT's downside sensitivity and uses DeepSeek's strength on mild positives without inflating extremes.

Reporting for decision-makers. Present a quarter-level 3-class summary (Positive/Neutral/Negative) for the executive view, with the 5-bucket breakdown and reliability notes in the appendix. Plot Manual, FinBERT, and DeepSeek quarter means on a common −2…+2 scale alongside the ±2-day market reaction, and track a small set of KPIs: Neutral share, Slight± shares, (Strong+ − Strong−) balance, model-agreement rate, and a count of tone–price divergences. This makes the signal legible at a glance and diagnostic when it matters.

Governance and quality. Sample-check 10–20% of "slight" paragraphs each quarter; re-benchmark twice per year against manual labels; and run sensitivity checks on the event window (±1 vs ±2 days) and return specification if required. Maintain probabilities, overrides, and reviewer notes for full auditability.

Conclusion. Used together with light oversight, FinBERT keeps the signal disciplined, the LLM keeps it responsive, and the combined view provides credible, business-ready insights that align with human judgement and market narratives. Crucially, the ensemble transforms sentiment from a noisy by-product into a leading indicator that can be governed, audited, and acted on quarter after quarter. The rule-set is simple

enough for production yet extensible, models can be upgraded without rewriting the playbook, and human review is reserved for the few genuinely ambiguous “slight” cases. When tone and price diverge, the gap becomes a prioritised investigation list, guidance language, surprise versus expectations, and valuation context, rather than an error to ignore. The design is portable beyond these three firms to other sectors and languages, provided the same paragraph-first discipline and periodic re-benchmarking are maintained. In short: a disciplined anchor, a momentum reader, and a light governance layer turn earnings-call text into an operator’s dashboard, one that flags downside early, surfaces upside credibly, and earns the trust of decision-makers.

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Appendices

Appendix A: LLM Prompt

You are a sentiment-analysis engine.

I will give you a JSON array of objects like:

[{ "id": 0, "text": "…"}, { "id": 1, "text": "…"}, …]

For each text, return a probability distribution over the five sentiment classes:

Strong Negative, Slightly Negative, Neutral, Slightly Positive, Strong Positive.

Label your columns:

id,

LLM_sentiment,

LLM_pct_strong_positive,

LLM_pct_slightly_positive,

LLM_pct_neutral,

LLM_pct_slightly_negative,

LLM_pct_strong_negative

—with each LLM_pct_* a float from 0.0–1.0 summing to 1.0, one row per paragraph,

no markdown fences or commentary, pure CSV.

-for sentiment, choose one of the five classes.

Appendix B: Amazon Results

Table B.1: *Overall Metrics for Amazon- FinBERT*

Metric	Value
Accuracy	0.6783
Macro-F1	0.4516
Weighted-F1	0.6435
Precision (Macro)	0.4663
Recall (Macro)	0.4989

Table B.2: *Amazon / FinBERT: Classification report (paragraph level)*

Class / Summary	Precision	Recall	F1-Score	Support
Strong Positive	0.714	1.000	0.833	10
Slightly Positive	0.722	0.271	0.394	48
Neutral	0.695	0.890	0.781	82
Slightly Negative	0.200	0.333	0.250	3
Strong Negative	0.000	0.000	0.000	0
accuracy	0.678	0.678	0.678	0.678
macro avg	0.466	0.499	0.452	143
weighted avg	0.695	0.678	0.643	143

Table B.3: *Amazon / FinBERT: Per-quarter performance*

Quarter	n	Accuracy	Macro-F1	Weighted-F1	Precision (Macro)	Recall (Macro)
Q1_2023	17	0.588	0.293	0.585	0.333	0.283
Q1_2024	18	0.667	0.590	0.581	0.556	0.636
Q2_2023	18	0.667	0.509	0.623	0.661	0.517
Q2_2024	19	0.895	0.929	0.889	0.962	0.917
Q3_2023	16	0.563	0.157	0.538	0.150	0.164
Q3_2024	19	0.789	0.811	0.777	0.840	0.803
Q4_2023	19	0.632	0.540	0.631	0.559	0.538
Q4_2024	17	0.588	0.354	0.436	0.275	0.500

Table B.4: *Overall Metrics for Amazon- DeepSeek*

Metric	Value
Accuracy	0.6853
Macro-F1	0.3564
Weighted-F1	0.6576
Precision (Macro)	0.3422
Recall (Macro)	0.3852

Table B.5: *Amazon / DeepSeek: Classification report (paragraph level)*

Class / Summary	Precision	Recall	F1-Score	Support
Strong Positive	0.000	0.000	0.000	10

Slightly Positive	0.563	0.833	0.672	48
Neutral	0.806	0.707	0.753	82
Slightly Negative	0.000	0.000	0.000	3
Strong Negative	0.000	0.000	0.000	0
accuracy	0.685	0.685	0.685	0.685
macro avg	0.274	0.308	0.285	143
weighted avg	0.651	0.685	0.658	143

Table B.6: *Amazon / FinBERT: Per-quarter performance*

Quarter	n	Accuracy	Macro-F1	Weighted-F1	Precision (Macro)	Recall (Macro)
Q1_2023	17	0.882	0.608	0.855	0.586	0.633
Q1_2024	18	0.778	0.542	0.737	0.508	0.606
Q2_2023	18	0.500	0.343	0.476	0.325	0.367
Q2_2024	19	0.579	0.289	0.548	0.274	0.307
Q3_2023	16	0.625	0.625	0.625	0.727	0.727
Q3_2024	19	0.684	0.475	0.661	0.463	0.520
Q4_2023	19	0.684	0.481	0.645	0.524	0.519
Q4_2024	17	0.765	0.408	0.718	0.390	0.431

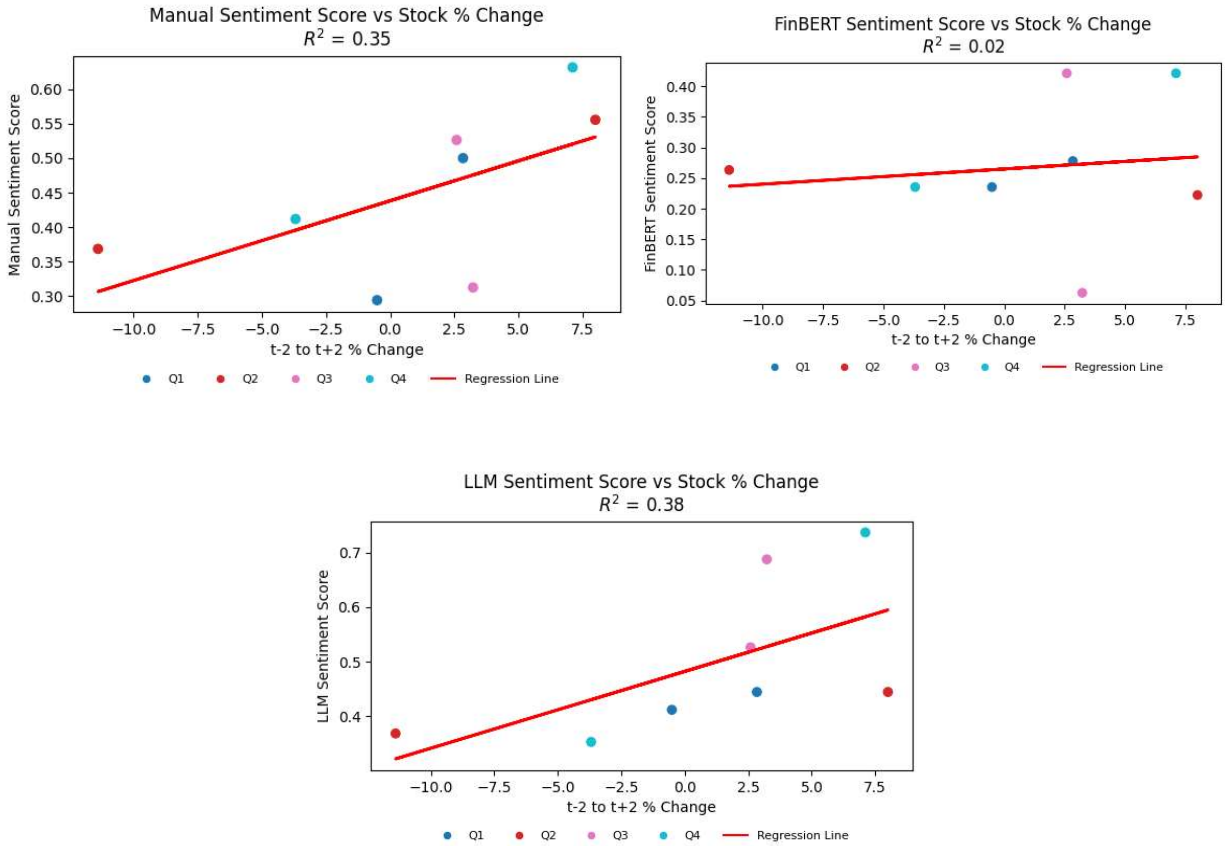

Figure B.1: *Sentiment Score against Stock Percentage Change*

Appendix C: Apple Results

Table C.1: *Overall Metrics for Apple- FinBERT*

Metric	Value
Accuracy	0.7430
Macro-F1	0.4162
Weighted-F1	0.7442
Precision (Macro)	0.4344
Recall (Macro)	0.4013

Table C.2: *Apple / FinBERT: Classification report (paragraph level)*

Class / Summary	Precision	Recall	F1-Score	Support
Strong Positive	0.464	0.371	0.413	35
Slightly Positive	0.396	0.380	0.388	50
Neutral	0.857	0.871	0.864	317
Slightly Negative	0.455	0.385	0.417	26
Strong Negative	0.000	0.000	0.000	0
accuracy	0.743	0.743	0.743	0.743
macro avg	0.434	0.401	0.416	428
weighted avg	0.747	0.743	0.744	428

Table C.3: *Apple / FinBERT: Per-quarter performance*

Quarter	n	Accuracy	Macro-F1	Weighted-F1	Precision (Macro)	Recall (Macro)
Q4_2021	59	0.949	0.874	0.955	0.813	0.986

Q1_2022	52	0.904	0.838	0.904	0.817	0.864
Q2_2022	56	0.768	0.333	0.784	0.355	0.441
Q3_2022	46	0.826	0.700	0.824	0.685	0.741
Q4_2022	45	0.622	0.281	0.632	0.330	0.257
Q1_2023	52	0.500	0.166	0.451	0.162	0.179
Q4_2024	57	0.719	0.343	0.693	0.450	0.358
Q1_2025	61	0.639	0.251	0.638	0.259	0.246

Table C.4: *Overall Metrics for Apple- DeepSeek*

Metric	**Value**
Accuracy	0.7150
Macro-F1	0.3756
Weighted-F1	0.6977
Precision (Macro)	0.6045
Recall (Macro)	0.4124

Table C.5: *Apple / DeepSeek: Classification report (paragraph level)*

Class / Summary	**Precision**	**Recall**	**F1-Score**	**Support**
Strong Positive	0.750	0.086	0.154	35
Slightly Positive	0.309	0.680	0.425	50
Neutral	0.859	0.845	0.852	317
Slightly Negative	0.500	0.038	0.071	26

Strong Negative	0.000	0.000	0.000	0
accuracy	0.715	0.715	0.715	428
macro avg	0.484	0.330	0.300	428
weighted avg	0.764	0.715	0.698	428

Table C.6: *Apple / DeepSeek: Per-quarter performance*

Quarter	n	Accuracy	Macro-F1	Weighted-F1	Precision (Macro)	Recall (Macro)
Q4_2021	59	0.814	0.301	0.825	0.285	0.388
Q1_2022	52	0.750	0.587	0.788	0.740	0.646
Q2_2022	56	0.714	0.298	0.710	0.274	0.451
Q3_2022	46	0.630	0.284	0.577	0.245	0.371
Q4_2022	45	0.667	0.273	0.609	0.252	0.297
Q1_2023	52	0.596	0.310	0.552	0.283	0.367
Q4_2024	57	0.772	0.508	0.744	0.811	0.480
Q1_2025	61	0.738	0.360	0.720	0.338	0.414

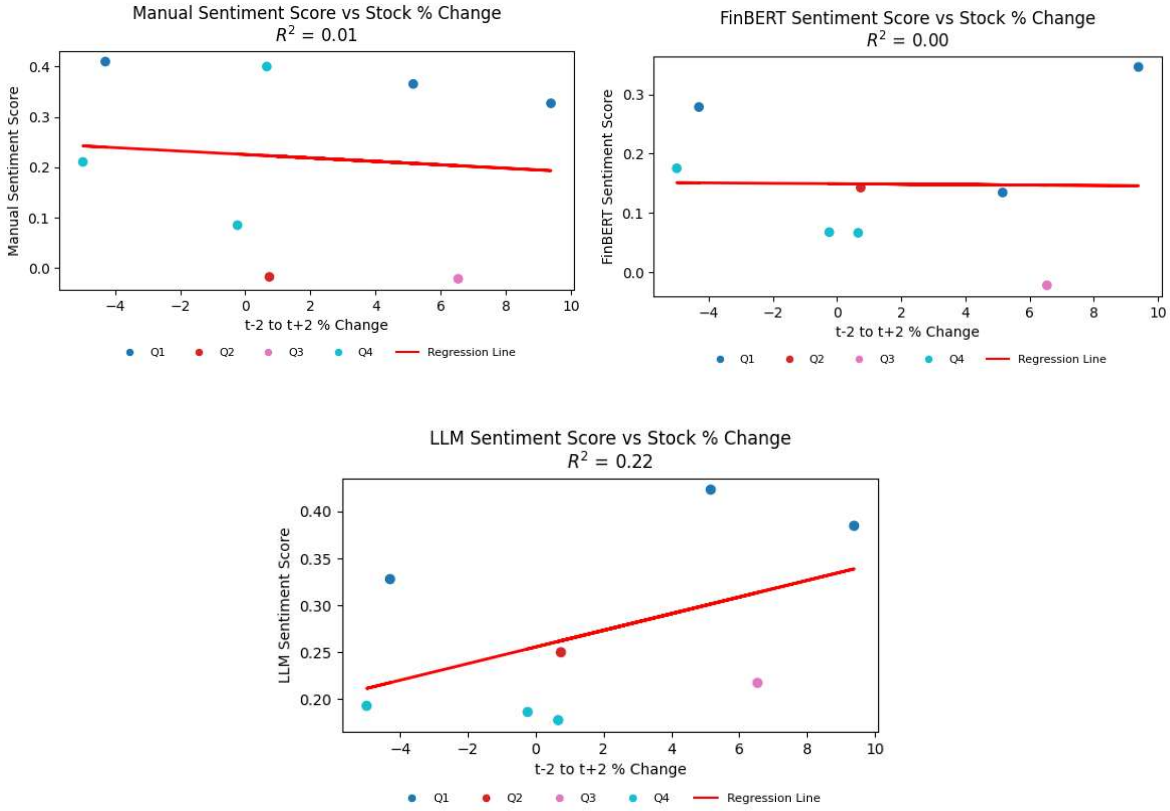

Figure C.1: *Sentiment Score against Stock Percentage Change*

Appendix D: NVIDIA Results

Table D.1: *Overall Metrics for NVIDIA- FinBERT*

Metric	Value
Accuracy	0.5771
Macro-F1	0.3103
Weighted-F1	0.4851
Precision (Macro)	0.5551
Recall (Macro)	0.2982

Table D.2: *NVIDIA / FinBERT: Classification report (paragraph level)*

Class / Summary	Precision	Recall	F1-Score	Support
Strong Positive	0.700	0.132	0.222	53
Slightly Positive	0.500	0.128	0.204	39
Neutral	0.575	0.981	0.725	105
Slightly Negative	1.000	0.250	0.400	4
Strong Negative	0.000	0.000	0.000	0
accuracy	0.577	0.577	0.577	0.577
macro avg	0.555	0.298	0.310	201
weighted avg	0.602	0.577	0.485	201

Table D.3: *NVIDIA / FinBERT:* Per-*quarter performance*

Quarter	n	Accuracy	Macro-F1	Weighted-F1	Precision (Macro)	Recall (Macro)
Q1_2024	26	0.769	0.607	0.727	0.913	0.550
Q1_2025	30	0.633	0.477	0.561	0.864	0.472
Q2_2024	25	0.560	0.336	0.433	0.355	0.400
Q2_2025	22	0.545	0.313	0.423	0.508	0.381
Q3_2024	21	0.476	0.292	0.357	0.375	0.333
Q3_2025	27	0.556	0.229	0.466	0.396	0.261
Q4_2023	26	0.423	0.251	0.334	0.300	0.310
Q4_2025	24	0.625	0.363	0.511	0.536	0.400

Table D.4: *Overall Metrics for NVIDIA – DeepSeek*

Metric	Value
Accuracy	0.5622
Macro-F1	0.5667
Weighted-F1	0.5600
Precision (Macro)	0.7824
Recall (Macro)	0.5826

Table D.5: *NVIDIA / DeepSeek: Classification report (paragraph level)*

Class / Summary	Precision	Recall	F1-Score	Support
Strong Positive	1.000	0.132	0.233	53
Slightly Positive	0.279	0.744	0.406	39
Neutral	0.851	0.705	0.771	105
Slightly Negative	1.000	0.750	0.857	4
Strong Negative	0.000	0.000	0.000	0
accuracy	0.562	0.562	0.562	0.562
macro avg	0.626	0.466	0.453	201
weighted avg	0.782	0.562	0.560	201

Table D.6: *NVIDIA / DeepSeek: Per-quarter performance*

Quarter	n	Accuracy	Macro-F1	Weighted-F1	Precision (Macro)	Recall (Macro)
Q1_2024	**26**	0.500	0.473	0.527	0.759	0.554
Q1_2025	**30**	0.567	0.372	0.511	0.337	0.417
Q2_2024	**25**	0.400	0.361	0.341	0.773	0.451
Q2_2025	**22**	0.500	0.322	0.430	0.280	0.386
Q3_2024	**21**	0.762	0.750	0.778	0.806	0.764
Q3_2025	**27**	0.556	0.403	0.515	0.379	0.600
Q4_2023	**26**	0.577	0.391	0.505	0.549	0.523
Q4_2025	**24**	0.667	0.560	0.667	0.760	0.595

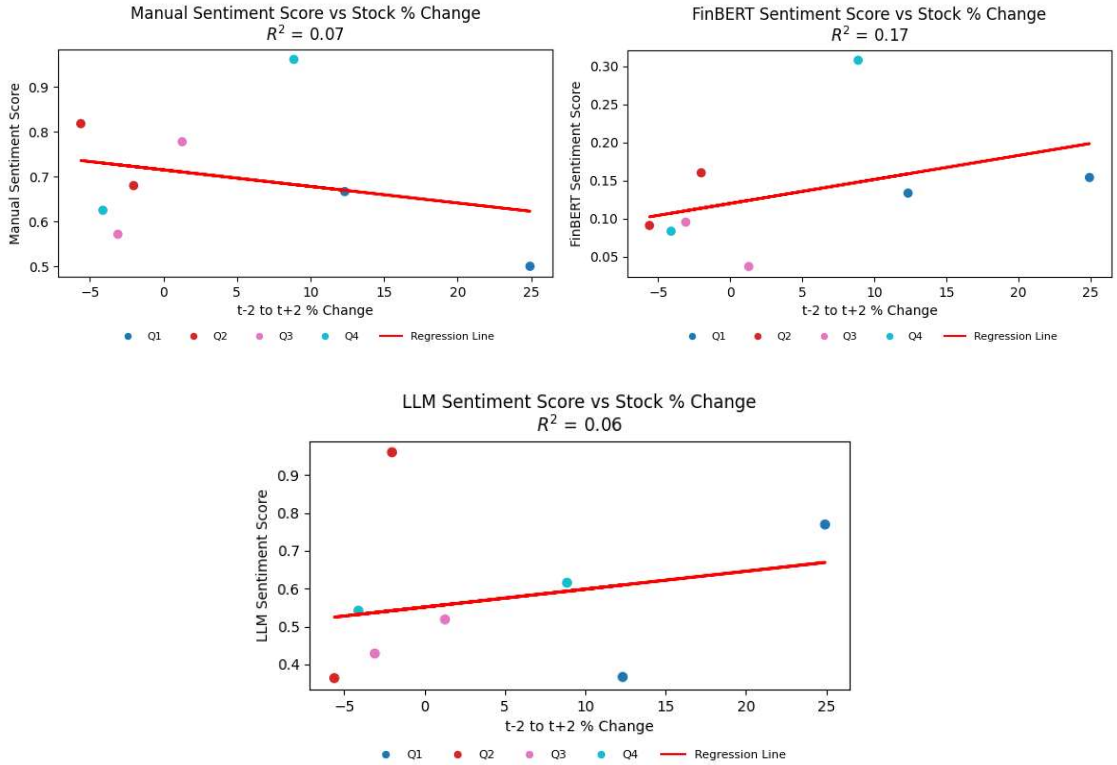

Figure D.1: *Sentiment Score against Stock Percentage Change*